

Personalization of Intelligent Virtual Agents for Motion Training in Social Settings

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Figure 1: Various roles of intelligent virtual agents—reluctant companion, therapist, or energetic trainer—in physical training scenarios.

ABSTRACT

Intelligent Virtual Agents (IVAs) have become ubiquitous in our daily lives, displaying increased complexity of form and function. Initial IVA development efforts provided basic functionality to suit users' needs, typically in work or educational settings, but are now present in numerous contexts in more realistic, complex forms. In this paper, we focus on personalization of embodied human intelligent virtual agents to assist individuals as part of physical training "exergames".

Index Terms: Human-centered computing—Intelligent virtual agents—Conversational agents—Human-centered computing—Ubiquitous and mobile computing—Ubiquitous and mobile devices—Personal digital assistants

1 INTRODUCTION

Initial efforts in Intelligent Virtual Agents (IVA) development focused on providing basic functionality to suit the needs of given contexts, however, IVAs of today are used in a wide range of applications, taking on increasingly realistic and/or complex forms. They assist us in our homes, typically as voice-based agents such as Amazon's Alexa, and entertain us in digital media such as video games, in the form of non-player characters, enriching our lives in increasingly many unexpected ways. In this paper we focus on embodied human intelligent virtual agents (IVAs) in social physical training contexts—in particular, how they may assist individuals as part of physical training through a variety of roles and variations in training approach. In this context, the degree to which the form and function, and especially the level of realism with which IVAs are

developed provides an important contribution to the overall ability of such systems to successfully help users to improve their health and overall well-being. Success of training systems relies on continued engagement of users, and as such the IVAs assisting in these systems must be well-suited to the task of conveying accurate information about exercises, providing feedback for user performance, and serving as surrogates in social activities such as group training scenarios. In this way, the refinement of embodied IVA qualities, for instance, regarding realism of appearance, knowledge, and behavior (both in level of detail, and inclusion of human-like imperfections), becomes of particular importance for health improvement systems, as detailed in this project.

Medical technologies pursue innovative solutions to modern therapeutic methods that provide improved access to information in a variety of contexts. In the field of general fitness, games providing engaging exercise possibilities (referred to as "exergames") are an increasingly popular focus of research. For many exergame training scenarios, IVAs serve as trainers or may provide social support in cases of group training or competition [9, 21]. As the fields of artificial intelligence (AI), virtual and augmented reality (VR/AR), etc. develop, they provide opportunities to further improve IVA traits and interaction capabilities [14]. The work described here is one facet of a project that incorporates gamification, motion recognition, and dynamic multimodal feedback, including interactions with humanoid IVA interactions. The findings should be generalized so as to support incorporation in typical exergaming contexts, including the possibility of VR/AR applications [5].

We consider the following IVA attributes and capabilities with a focus on how they might address the needs of users with regard to training optimization, healthy habit formation, as well as overall enjoyment and continued engagement:

- Realism and imperfections in IVAs, with a focus on behavior
- Affect and rapport in human-IVA interactions
- Personalization/adaptation of IVAs

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The remainder of this paper is structured as follows. Section 2 provides an overview of prior work in research and development of embodied IVAs in motion training settings, followed by a summary of related research projects with a focus on IVA realism. Section 2.1 is focused on requirements of IVAs in training contexts. In Section 2.2, we discuss aspects of training that contribute to users' success. As a basis for further discussion of IVA traits, Section 2.3 provides a brief overview of the roles IVAs might take to address the needs of users in training scenarios. The desirable IVA features, commonly present in human participants in training scenarios, which may be incorporated into embodied IVAs are also described. In Section 3, the methodology followed to create and evaluate our embodied IVAs is briefly described. The paper is concluded by Section 4, which summarizes the main aspects of the project and discusses our future developments and evaluation of IVAs for physical training.

2 RELATED WORK

In order for an exercise regimen to improve health outcomes, individuals must build and maintain the habit of regular physical activity. However, this is not often the case, especially with regard to people, who have not had a history of physical fitness practice. Accordingly, much research has explored the topic of motivations and adherence to training regimens, and factors that may increase the likelihood of users to do so [16].

During physical exercise training, information should be conveyed to provide accurate information about speed and positioning during initial movements, at minimum, as well as muscle activation and other physiological issues. After the initial demonstration, correction of movements requires the ability to focus on the most critical aspects in need of attention in order to reduce the cognitive load of beginners or to help more advanced users focus on aspects related to their personal goals [18]. Virtual trainers must be able to perform these functions in a way that not only provides sound advice, but also does so in a manner that encourages users to continue with their training regimen.

2.1 Training with Virtual Agents

Extensive research and development has been conducted with the goal of creating IVAs for training, with a significant portion focused on realistic embodied humanoid IVAs for physical training purposes [5, 18]. This is primarily due to the need for "motivation and guidance" as they are "considered main barriers for people to overcome when trying to exercise regularly" [5]. This is especially true for those with mobility impairments, which becomes more prevalent in older populations. These issues are compounded by the monotony and isolation of in-home training. Some of the basic functions that IVA trainers can perform in such contexts include explanation and performance of exercises, monitoring and correcting users' practice, assessing users' physical and emotional states, tracking progress toward goals adjustment of exercises based on a combination of these factors, motivational user feedback, and adjusting their own traits and behaviors to improve the users' experience and overall affective state to provide an optimal experience for the user based on their background and goals. While some studies focus on adaptation of training systems to accommodate those who prefer competitive activities versus cooperative or solitary activities [21], this work does not venture into that space. The findings from this project's results may, however, provide benefit in development of robotic agents performing similar support roles in physical training scenarios [7].

2.2 Support Training Adherence through Engaging Interactions

2.2.1 Training and Motivation

Users of training applications must maintain continued use in order to benefit from them, however, people who initiate new training regimens often give up, for numerous reasons. Exercise adherence,

and the motivations (intrinsic and extrinsic) that support it, have been long-standing topics of research across the general population [16], and with increasing interest in older adults as a particular focus [15, 17]. While individuals may start training based on initial motivations such as enjoyment, sense of competence, body-related concerns, and social engagement, they might factor in new motives such as training atmosphere and progress toward goals, while they build and maintain their training habits. With the knowledge that it takes approximately ten weeks to build a stable habit, development of training systems and, in particular, IVA functionalities should stress minimization of habit-weakening factors (such as unnecessary variation) and reinforcement of strengthening factors (such as small, simple changes in behavior or use of memory aids) [6, 22].

2.2.2 Role of IVAs in Training and Motivation

Simple verbal feedback from a virtual coach has been shown to improve motivation in the short term [4], but further studies would be needed to determine effects on motivation over the course of a long term training routine. In another study, the support of a non-humanoid virtual trainer was found to be engaging [1], but users were frustrated when the sensor-based system failed to recognize that they had reached their daily activity goals, instead providing negative feedback. The precise role the IVA plays within the activity can also have varying effects. Depending on whether the IVA behaves in a competitive or cooperative manner, the user might feel motivated to exercise for longer, but this feeling may not correlate to greater enjoyment of the activity [20], which may then hinder longer term motivation to continue with the training regimen. A friendly, personable virtual trainer may improve the training experience for older individuals who may not have easy access to an in-home personal trainer [13]. Such an interplay of competing factors makes it clear that a carefully crafted balance of roles should be considered to provide an optimal training experience for a variety of workout styles.

2.3 Embodied IVAs Traits in Motion Training Scenarios

2.3.1 (Behavioral) Realism and Imperfections in IVAs

In order for users to receive the best possible training experience, they must not only enjoy the exercise activities, but also trust that the information the IVA trainer (or 'virtual trainer') provides will be accurate. To support this sense of believability in the virtual trainer's ability, and ultimately build a rapport between the trainer and user, attention to detail with regard to levels of realism in design of such trainers is important. This simulation would extend not only to realistic human traits and behaviors, but also to emotional states and physiological responses that may be observed by users. An example might be that the trainer also shows signs of sweating during extended exertion, which a user might find more relatable.

When possible, it is best to observe human interactions to provide a basis for emulation. The ERISA framework uses recorded human activity data to create emotionally realistic social training companions [3]. By incorporating complex realistic gesture and facial action units (FACs), and recognition of six basic emotions they endeavor to enhance the believability and social connectedness of users and their agents. Through automatic behavior modeling, they provide their companions with 'Memories' (internal states), 'Personalities', and 'Social Relationship Models'. Based on their Smile Game evaluation, these efforts resulted in limited success, with the explanation that the Speech Recognition module lowered the overall performance of the system. Motion capture recording technologies provide the ability to replicate motions such as head movements, facial expressions, and hand gestures during conversations, as well as full body motion. Reliance on realistic modes of nonverbal communication can be approached through motion capture recording of human conversations, as with the MSP-AVATAR Corpus [19]. As so much of human discourse is conveyed through nonverbal communication



Figure 2: Realistic virtual trainer exhibiting perspiration and disheveled hair. Created using CC3 and IC7.

(hand and body position, facial expression, etc) as well as concurrent verbal exchanges, direct recording provides a rich resource to build realistic interaction models upon when attempting to synthesize IVA behaviors. Similar systems have been developed to create interview functionality for virtual health assistants [11], albeit through use of microphone and video camera recording of expressions only. Special attention has been given to co-verbal gesture synchronicity [10], since the natural flow of conversation is an important component of training interactions. In particular, a trainer's ability to provide information about an exercise and gesture toward the part of the body in motion while attempting to convey correct position, form, and speed is of utmost importance.

Other approaches to increasing the realism and, consequently, the believability of IVAs might be to incorporate human-like episodic memory [2], or algorithms to generate behaviors for individual IVAs [12], which may form the basis for sufficiently advanced agents that might one day pass the Turing test.

2.3.2 Rapport and Cultivation of Motivating Affective Responses in Human-IVA Interactions through Personalization

With the understanding that positive affect regarding physical activity sessions may predict further activity levels [23], it is clear that maintaining an enjoyable training atmosphere as well as a good relationship or overall rapport between the virtual trainer and the user is an important goal for this project. With this in mind, we must also be aware that effects of the interplay between each individual users' personality and those of the IVAs we develop may lead to highly variable outcome, in terms of their level of trust [8], and overall enjoyment of the training program.

3 METHODOLOGIES

For this paper, we have approached the development of realistic IVAs from the position that users would often prefer to workout in a social setting, but may not have the opportunity due to a variety of reasons—with maintenance of strict hygiene standards being chief among them during the ongoing COVID-19 pandemic. With this in mind, we develop a variety of virtual trainer and virtual training companion personas, that users may interact with during training activities. The basic categories, and further refinement of these

personas is best based on feedback from potential users. In this way, we can determine the level of realism, degrees to which each IVA persona may vary, and the overall attitude that may be displayed by each IVA so that it might complement the training style of potential users.

Personalization of IVA Trainer For virtual trainers, the appearance of competence and expertise in their given field is of great importance. However, we must strive to develop agents whose physical qualities convey an accurate representation of their technical capabilities. As an example, the realistic representation of a trainer as displayed in Figure 2, may prompt a greater expectation of human capability than that of a more cartoonish or non-humanoid IVA. This balance is important because any mismatch in the ability of the IVA to correctly react to the user may result in a loss of trust in the trainer, and may reduce the level of enjoyment and engagement with which they interact with the system.

Personalization of IVA Training Companion Similar to virtual trainers, the development of virtual training companions must take into account the wide variations in user personalities, training motivations, and ongoing training preferences. Users may prefer to train with one type of companion versus another in a given context. They may also wish to adjust the energy of their companions to suit their motivational needs on a given day, for example through personality gradients (as shown in Figure 1). In the image, the agents display idle poses and expressions meant to convey varying degrees of 'attitude', including emotional valence, motivation, and general energy level. The Intensity slider shown underneath is a direct manner of manipulating these qualities, but more sudden modes of personalization, e.g. through interaction with the system, may result in a more engaging experience. Users' ability to participate with virtual training companions and virtual trainers simultaneously also provides the opportunity for them to benefit from interactions between trainers and companions. In this way, a trainer might perform an exercise with the ideal form, and then provide posture feedback to a companion that may perform a slight variation due to a difference in fitness level or potential mobility issue, as an example. Such and interaction provides a view of the activity from another perspective, and might also be more amenable to certain individuals, in cases where they are not comfortable with being critiqued personally.

Evaluation of IVAs Preliminary study of the effects of varied levels of realism would address simple changes in IVA appearance, i.e., three levels of model detail/realism. A secondary evaluation would focus on provision of a means of direct personalization of IVAs qualities by the user, with data collected regarding preferences. Further evaluation could then entail recommended adjustment of IVA appearance and/or behaviour with the intent of suiting a users' potential preferences based on their previous physical fitness background and current fitness goals. Conducting such studies is difficult during the COVID-19 pandemic, but substitution of in-person evaluations with remote testing through VR systems is a viable option. In this way, we would be able to present more varied options to a wider range of potential users (with the limiting factors being availability and access to head-mounted displays), and iterate on feedback in a shorter span of time without as great need for strict hygiene standards due to in-person studies.

4 CONCLUSION AND FUTURE WORK

The broad array of opportunities to develop and refine IVAs in physical training scenarios provides exciting avenues to explore the interplay of motivational approaches that may benefit users. Further, there are interesting challenges to address, such as balancing levels of realism and typical human imperfections with ideal qualities, in the pursuit of exceptional training experiences. As our work in this paper progresses, we will take advantage of opportunities in remote

VR testing to gain a greater understanding of the needs of users with regard to IVA interactions in social training settings.

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